

Aurélien Cécille

cecille.aurelien@iris.cnrs.fr

+33 6 09 22 03 96

7 rue des Jardins, 69600 Oullins, France

Aurel-C

Education

Ph.D. Candidate in Computer Science

LIRIS & Visual Behavior

May 2023 - Present

Structure Aware Self-Supervised Monocular Depth Estimation

Master's Degree in Computer Science

University of Technology of Troyes

2017 - 2022

Work Experience

Research Engineer

Visual Behavior

Oct. 2022 - May 2023

- Research, training, and evaluation of deep learning models for robotic and intelligent vehicle vision

AI Engineer Intern

Orange

Feb. 2022 - Jul. 2022

- Detection and segmentation of fiber optic cables in images
- Few-shot learning methods based on meta-learning
- Leveraging latent representations from generative models (GANs)

AI Engineer Intern

Visual Behavior

Sept. 2020 - Feb. 2021

- Self-supervised learning for depth estimation, optical flow, and egomotion estimation
- Environment reconstruction from bird's-eye view

Projects

Conditional Logo Generation

- Logo generation using StyleGAN2
- Conditioning the generator with latent-space clustering from a VAE
- Flask interface with color, shape, and category controls

Reinforcement Learning Agent

- Custom grid-world environment in Unity ML-Agents linked with OpenAI Gym
- Implementation in JAX for parallelization and optimized hyperparameter search

Research Interests

Computer Vision

Deep Learning

Self-Supervised Learning

Summer Schools & Workshops

ICVSS 2025

Computer Vision for Spatial Intelligence

Summer school attendance & poster

PAISS 2025

Summer school attendance

Naver Labs Workshop 2025

AI for Robotics

Attendance and poster presentation

Publications

Towards Sharper Object Boundaries in Self-Supervised Depth Estimation

Aurélien Cécille, Stefan Duffner, Franck Davoine, Rémi Agier & Thibault Neveu

BMVC oral, November 27, 2025, Sheffield, United Kingdom

GroCo: Ground Constraint for Metric Self-Supervised Monocular Depth

Aurélien Cécille, Stefan Duffner, Franck Davoine, Thibault Neveu & Rémi Agier

ECCV 2024, October 2024, Milan, Italy

Languages

English : Fluent (C2 level)

French : Native language